

# WMC\_Ver01\_CoreCycle\_Requirements

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## Requirement Set: WMC\_Ver01\_CoreCycle\_Requirements

### Description

**Change Information**    No change issue detected.

## 1 System Initialization

**Requirement Type** Functional

**ID** WMC-REQ-001

**Description**

On System start, the controller shall enter the Idle stage with all actuators OFF, cycle\_complete set to 0, and fault\_indicator set to 0.

**Change Information** No change issue detected.

**Links**

**Artifact** [WMC\\_Ver\\_01\\_CoreCycle.slx](#)

 [Idle](#) (↔Implemented by )

## 2 Start Condition

**Requirement Type** Functional

**ID** WMC-REQ-002

**Description**

When start\_button is pressed and door\_closed is 1, the controller shall start the washing cycle.

**Change Information** No change issue detected.

**Links**

**Artifact** [WMC\\_Ver\\_01\\_CoreCycle.slx](#)

 [\[start\\_button == 1 && door\\_closed == 1\]](#) (↔Implemented by )

## 3 Door Safety Before Start

**Requirement Type** Functional

**ID** WMC-REQ-003

**Description**

If start\_button is pressed while door\_closed is 0, the controller shall remain in Idle state and not initiate the cycle.

**Change Information** No change issue detected.

**Links**

**Artifact** [WMC\\_Ver\\_01\\_CoreCycle.slx](#)

 [\[door\\_closed == 0\]](#) (↔Implemented by )

## 4 Door Lock Activation

**Requirement Type** Functional

**ID** WMC-REQ-004

**Description**

After a valid start command, the controller shall activate door\_lock and shall wait for door\_lock\_delay duration before entering the Water\_Fill\_Wash\_Cycle stage.

**Change Information** No change issue detected.

**Links**

**Artifact** [WMC\\_Ver\\_01\\_CoreCycle.slx](#)

 [Door Lock Activation](#) (↔Implemented by )

 [\[after\(door\\_lock\\_delay, sec\) && door\\_closed ==1\]](#) (  Implemented by )

## 5 Wash-Cycle Water Fill

**Requirement Type** Functional

**ID** WMC-REQ-005

### Description

During Water\_Fill\_Wash\_Cycle stage, the controller shall turn ON water\_valve and shall remain in this stage until water\_level\_ok becomes 1. The water\_level\_ok is shared with the rinse-cycle water fill stage.

**Change Information** No change issue detected.

### Links

**Artifact** [WMC Ver 01 CoreCycle.slx](#)

 [Water Fill Wash Cycle](#) (  Implemented by )

 [\[water\\_level\\_ok == 1\]](#) (  Implemented by )

## 6 Detergent Fill Stage

**Requirement Type** Functional

**ID** WMC-REQ-006

### Description

After wash-cycle water filling is completed, the controller shall enter Detergent\_Fill stage and turn ON detergent\_valve until detergent\_level\_ok becomes 1.

**Change Information** No change issue detected.

### Links

**Artifact** [WMC Ver 01 CoreCycle.slx](#)

 [Detergent Fill](#) (  Implemented by )

 [\[detergent\\_level\\_ok == 1\]](#) (  Implemented by )

## 7 Wash Stage

**Requirement Type** Functional

**ID** WMC-REQ-007

### Description

After detergent filling is completed, the controller shall enter the Wash stage and turn on wash\_motor for the configured wash\_duration.

**Change Information** No change issue detected.

### Links

**Artifact** [WMC Ver 01 CoreCycle.slx](#)

 [Wash](#) (  Implemented by )

 [\[after\(wash\\_duration, sec\)\]](#) (  Implemented by )

## 8 Wash-Cycle Drain

**Requirement Type** Functional

**ID** WMC-REQ-008

### Description

After wash\_duration is completed, the controller shall enter Drain\_Wash\_Cycle stage and turn ON drain\_pump for the configured drain\_duration. The drain\_duration parameter is shared with the Drain\_Rinse\_Cycle stage.

**Change Information** No change issue detected.

#### Links

**Artifact** [WMC Ver 01 CoreCycle.slx](#)

 [Drain Wash Cycle](#) (↔Implemented by )

 [\[after\(drain\\_duration, sec\)\]](#) (↔Implemented by )

## 9 Rinse-Cycle Water Fill

**Requirement Type** Functional

**ID** WMC-REQ-009

#### Description

After wash-cycle draining is completed, the controller shall enter Water\_Fill\_Rinse\_Cycle stage and shall turn ON water\_valve until water\_level\_ok become 1. The water\_level\_ok signal is shared with the wash-cycle water fill stage.

**Change Information** No change issue detected.

#### Links

**Artifact** [WMC Ver 01 CoreCycle.slx](#)

 [Water Fill Rinse Cycle](#) (↔Implemented by )

 [\[water\\_level\\_ok == 1\]](#) (↔Implemented by )

## 10 Rinse Stage

**Requirement Type** Functional

**ID** WMC-REQ-010

#### Description

After rinse-cycle water filling is completed, the controller shall enter the Rinse stage and turn ON wash\_motor for the configured rinse\_duration.

**Change Information** No change issue detected.

#### Links

**Artifact** [WMC Ver 01 CoreCycle.slx](#)

 [Rinse](#) (↔Implemented by )

 [\[after\(rinse\\_duration, sec\)\]](#) (↔Implemented by )

## 11 Rinse-Cycle Drain

**Requirement Type** Functional

**ID** WMC-REQ-011

#### Description

After rinse\_duration is completed, the controller shall enter Drain\_Rinse\_Cycle stage and turn ON drain\_pump for the configured drain\_duration. The drain\_duration parameter is shared with the Drain\_Wash\_Cycle stage.

**Change Information** No change issue detected.

#### Links

**Artifact**     [WMC Ver 01\\_CoreCycle.slx](#)

 [Drain Rinse Cycle](#) (↔Implemented by )

 [\[after\(drain\\_duration, sec\)\]](#) (↔Implemented by )

## 12 Spin Stage

**Requirement Type**     Functional

**ID**     WMC-REQ-012

### Description

After rinse-cycle draining is completed, the controller shall enter Spin stage and turn ON the spin\_motor for the configured spin\_duration.

**Change Information**     No change issue detected.

### Links

**Artifact**     [WMC Ver 01\\_CoreCycle.slx](#)

 [Spin](#) (↔Implemented by )

 [\[after\(spin\\_duration, sec\)\]](#) (↔Implemented by )

## 13 Cycle Completion

**Requirement Type**     Functional

**ID**     WMC-REQ-013

### Description

After spin\_duration is completed, the controller shall enter Cycle\_Complete stage, turn OFF all actuators, shall unlock the door\_lock, and shall set cycle\_complete to 1.

**Change Information**     No change issue detected.

### Links

**Artifact**     [WMC Ver 01\\_CoreCycle.slx](#)

 [Cycle Complete](#) (↔Implemented by )

## Appendix

### Artifact List

Simulink models:

| # | Name                     | Folder                                                                                                              | Version |
|---|--------------------------|---------------------------------------------------------------------------------------------------------------------|---------|
| 1 | WMC_Ver_01_CoreCycle.slx | D:\MBD-SILMIL_May_2026\MBD-Requirements_Based_Stateflow_Projects\Washing Machine Project\WMC_Ver01_CoreCycle\models | 2.16    |